



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2014 – 07



MARIÑAS Destoner

MARIÑAS DESTONER

Test Report No.	:	2014 - 07
Date of Test	:	January 21, 2014
Test Valid Until	:	January 21, 2019
Brand	:	MARIÑAS
Model	:	-
Type	:	Specific gravity type
Manufacturer	:	MARIÑAS TECHNOLOGIES, INC. 260 Rizal St., Pila, Laguna
Test Requested by	:	METALS INDUSTRY RESEARCH AND DEVELOPMENT CENTER MIRDC Compound, Gen. Santos Ave., Taguig City, Metro Manila

SUMMARY

The MARIÑAS Destoner was tested using hulled paddy (brown rice) manually mixed with stones whose weight of 0.01% of the weight of the hulled paddy and with size ranging from 3.35 mm to 4.0 mm.

The MARIÑAS Destoner had an average Input Capacity of 1151 kg/h of hulled paddy destoner Shaft Speed of 1278 rpm. The destoner had Machine Efficiency of 90.5% of removing stones from the input material.

The Power Consumption of the electric motor which drove the destoner shaft was 0.31 kW, with Load Current of 1.95 amperes and Line Voltage of 229 volts.

One person was needed to operate the machine.

PURPOSE AND SCOPE OF TEST

The machine was tested to evaluate its performance using hulled paddy mixed with 0.01 % by weight of stone with size ranging from 3.35 mm to 4.0 mm. The destoner was tested at the Philippine Center for Postharvest Development and Mechanization (PhilMech) in Muñoz, Nueva Ecija on January 21, 2014. The test was carried out to assess and verify the following:

1. Input capacity
2. Machine efficiency to remove stone
3. Speed of components
4. Power Consumption
5. Labor requirement

OPERATOR'S MANUAL

No operator's manual or brochure was submitted.

DESCRIPTION OF THE MACHINE

The MARINAS Destoner as shown in **Figure 1**, was specific gravity type destoner. It was one of the components of a rice mill whose main function is to remove the stones that mixed with the paddy during milling operation. It was powered by a 0.37 kW electric motor which drove directly the destoner's shaft through a belt-pulley transmission system (**Fig. 2**). The machine consisted of the hopper, a vibrating screen and outlet chutes for stone and hulled paddy.

The hopper (**Fig. 3**) was where the brown rice was fed. It was made of metal sheet and installed above the vibrating screen.

An inclined vibrating screen with slotted perforations (**Fig. 4**) was installed inside the destoner housing. The stones that were mixed with the brown rice were separated utilizing the specific gravity separation principle. An outlet chute for stone (**Fig. 5**) was provided at one side of the destoner opposite the outlet chute for brown rice (**Fig. 6**).



Figure 1. MARINAS Destoner



Figure 2. Belt-pulley transmission system



Figure 3. Hopper

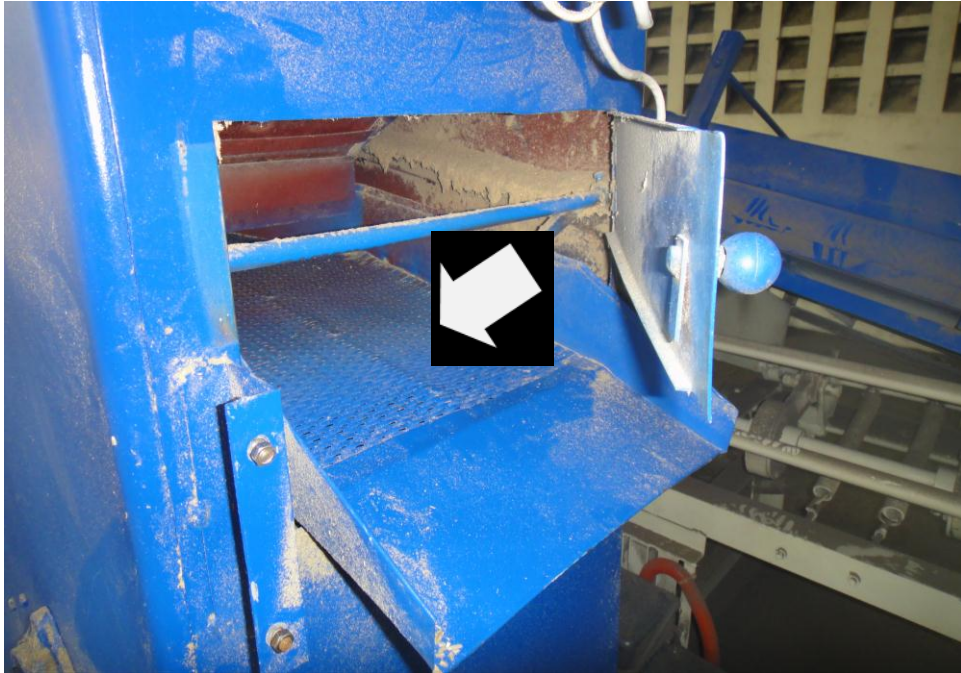


Figure 4. Vibrating Screen

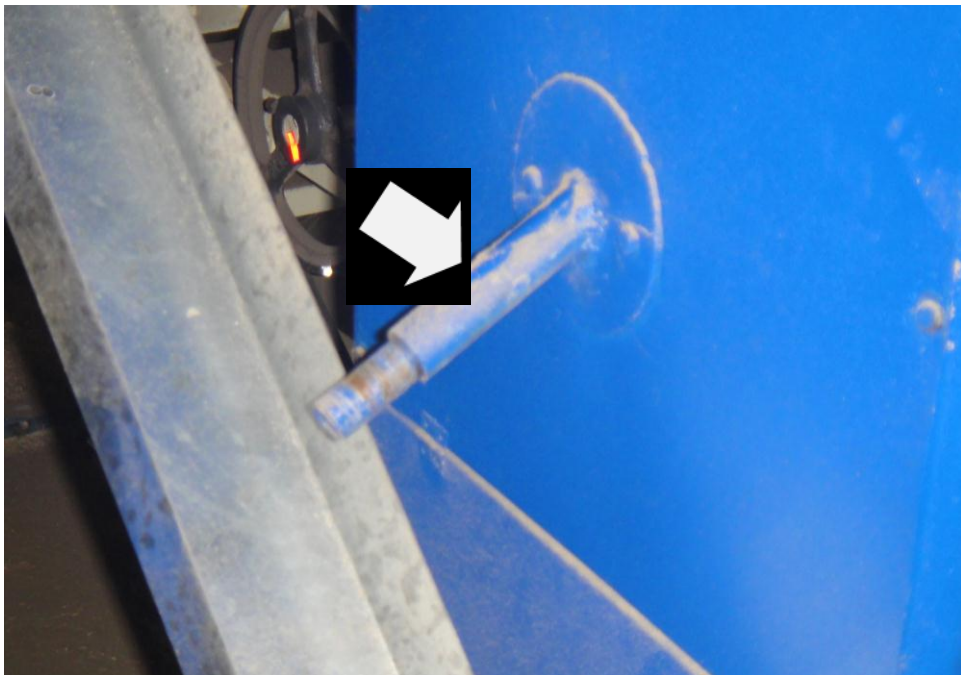


Figure 5. Outlet Chute for Stones



Figure 6. Outlet chute for brown rice

RESULTS AND DISCUSSIONS

The MARIÑAS Destoner was tested using hulled paddy (brown rice) manually mixed with stones whose weight is 0.01% of the weight of the hulled paddy. The stones passed thru the Tyler Standard Sieve and had sizes ranging from 3.35 mm to 4.0 mm. After passing thru the destoner, the weight of the stones that were removed was taken and recorded.

Results of performance tests are shown in **Table 2**.

Input Capacity

The Input Capacity of the destoner refers to the amount of material fed into the machine per unit of time. The machine had an average Input Capacity of 1151 kg/h.

Machine Efficiency (in removing stones)

This is the ratio of the weight of the stones that was not separated from the hulled paddy after passing the destoner and to the total amount of stones mixed with the hulled paddy before passing the destoner, expressed in percentage. The machine had a Machine Efficiency of 90.5% of removing stones.

Power Consumption

The average Power Consumption of the 0.37 kW electric motor was 0.31 kW with Load Current of 1.95 amperes and Line Voltage of 229 volts.

Labor Requirement

One person was needed to operate the destoner.

Observations

1. The outlet chute for the stones that were separated from the hulled paddy was not functional, i. e. it was blocked and the operator had to manually remove the stones from the vibrating screen after operation (**Figure 7**).
2. The belt-pulley transmission system of the destoner was not provided with cover (**Figure 8**).

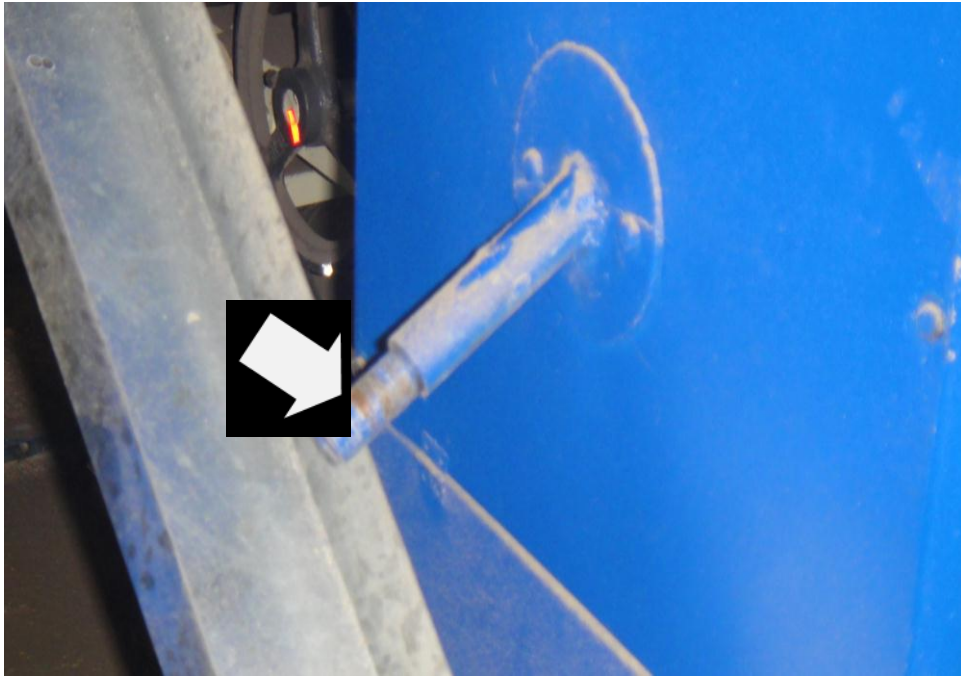


Figure 7. Outlet chute for stones was blocked



Figure 8. No belt cover

PERFORMANCE TEST

Machine Tested : MARINAS Destoner
(powered by an Electric Motor)

Date of Test : January 21, 2014

Location of Test : Philippine Center for Postharvest Development
and Mechanization (PhilMech)
Muñoz, Nueva Ecija

Table 1. Conditions of test material

ITEMS	
1. Input Material	Hulled paddy (brown rice) manually mix with stones whose weight is 0.01 % of the weight of hulled paddy
2. Size of stone (mm)	3.35 mm – 4.0 mm

Table 2. Result of Performance Test

ITEMS	Data
1. Weight of hulled paddy (brown rice) with stone, kg	126.6
2. Input Time, h	0.11
3. Input Capacity, kg/h	1151
4. Weight of stone present in the input material (kg)	1.27
5. Weight of stone removed (kg)	1.15
6. Machine's Efficiency to remove stone (%)	90.5

Table 2. Result of Performance Test of MARINAS Destoner (cont'n.)

ITEMS	Data
7. Speed of Components, rpm	
a) Electric Motor	
Without load	1760
With load	1759
b) Destoner Shaft	
Without load	1282
With load	1278
8. Power Consumption	
a) Power, kW	
Without load	0.30
With load	0.31
b) Voltage, V	
Without load	230
With load	229
c) Current, A	
Without load	1.85
With load	1.95

Test Engineer:

DARWIN C. ARANGUREN

Checked by:

ROMULO E. EUSEBIO

Approved for Release:

DELFIN C. SUMINISTRADO
Director

Date

N.B.

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